

HEMAL SHIRISH PATEL

12547 Highway 48, Menlo, GA 30731 | 706-936-4202 | U.S. Citizen
hsppatel12@gmail.com | [Portfolio](#) | [LinkedIn](#)

OBJECTIVE

FE-Certified Computer Engineer with hands-on experience testing embedded firmware, troubleshooting hardware-software systems, and bridging engineering and production teams — seeking to apply these strengths in a technical engineering role.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

2023

Bachelor of Science in Computer Engineering — GPA 3.06

Relevant Coursework: Embedded Systems Design, Real-Time Operating Systems, Processor Design (FPGA), Hardware/Software Systems (C/MIPS), FPGA-Based Digital Design (VHDL), VLSI, Machine Learning (Python), Cloud Computing (AWS)

TECHNICAL SKILLS

Embedded & Firmware: C, C++, MIPS Assembly, ARM mbed, PIC Microcontrollers, RTOS Fundamentals (task scheduling, priorities), SPI, I2C, UART

Hardware & Test Tools: Oscilloscopes, Digital Multimeters (DMM), Soldering, PCB Troubleshooting, Cadence Virtuoso, Quartus, Vivado, LTSpice, Athena

Programming & Cloud: Python, MATLAB, SQL, VHDL, Verilog, AWS Cloud Services

Other: FE Certified (EIT), Technical Documentation, English (Fluent), Gujarati (Conversational)

EXPERIENCE

Cennox — Alpharetta, GA

2025 – Present

Process / Product Engineer

- Stress-tested newly developed products by running repeated and edge-case scenarios on individual units to identify design and firmware failure points prior to release.
- Verified device-to-portal communication, documenting required software fixes for the management interface and tracking resolution with the development team.
- Performed root-cause analysis on vendor components and safe bodies by comparing physical units against CAD-derived build pack documentation, identifying damage and discrepancies against approved specifications.
- Escalated test findings and vendor issues directly to engineering leads and vendors, accelerating resolution of critical hardware and firmware defects.
- Authored SOPs adopted as training references for new hires and vendor partners, reducing onboarding ambiguity across the production floor.
- Supported testing and documentation across Smart Safe and Core product lines plus shipping/receiving, coordinating unit staging and deployment for enterprise banking and retail security clients.

BJ's Foodmart — Menlo, GA

2016 – Present

Operations Manager

- Managed facility operations, vendor coordination, and troubleshooting of on-site machinery and POS/network hardware, maintaining 100% uptime and regulatory compliance.

ENGINEERING PROJECTS

Smart Pipe System

Senior Design I & II

- Developed an end-to-end IoT monitoring system for water pressure and pH levels, using LoRa modules for long-range, low-power data transmission.
- Integrated hardware sensors with AWS Cloud to build a real-time notification system and remote monitoring dashboard, engineering the physical prototype with precision soldering and circuit assembly.

Tools: AWS, Arduino, LoRa, pH/Pressure Sensors, Soldering

Airplane Instrument Visualization

Embedded System Design (ECE 4180)

- Developed a real-time heads-up display (HUD) for pilots using an ARM mbed microcontroller, integrating IMU, GPS, and barometric sensors.
- Implemented multithreading and debugged sensor fusion issues to ensure accurate, high-frequency instrument data under real-time constraints.

IC Fabrication Lab

ECE 4452

- Fabricated integrated circuits in a cleanroom environment and performed post-fabrication reliability testing (Athena) to verify electrical performance against design specifications.